

# Stuti Thummala

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<https://stuti-product-design-portfolio.vercel.app/>

## EDUCATION

### Georgia Institute of Technology

December 2027

Bachelor of Science in **Computational Media (Computer Science + Interaction Design)** | Minor in **Business Administration**

- Relevant Coursework: Intro to Media Studies, Intro to AI, Algorithms, Data Structures, Objects & Design, Computer Organization & Programming (C/Linux), Software Development, Theory of Computing, Discrete Math, Applied Linear Algebra, Multivariable Calculus

### Certifications & Awards:

- Women in Technology (WIT) Girl of the Year Finalist — Selected as 1 of 3 finalists from a global network of 48,000+ women.
- GIAC Foundational Cybersecurity Technologies (GFACT) — August 2023 (Expires: August 2027)
- SANS SEC275: Computers, Technology, & Security — August 2023
- IT Specialist - Java (Certiport) — April 2024
- IT Specialist - Software Development (Certiport) — April 2024

## SKILLS

**Design & Prototyping:** Figma (Auto Layout, constraints, components, variants, interactive prototypes), FigJam, usability heuristics

**Frontend & Interaction:** React, Next.js, JavaScript, JavaFX (FXML + CSS), D3.js, Three.js, animation systems, component-based UI design

**Languages:** Python, Java, C, JavaScript, MATLAB

**Cloud/DevOps:** AWS (Lambda, API Gateway, DynamoDB, S3, IAM, CloudWatch, SNS/SQS), Docker, Terraform, GitHub Actions

**Frameworks/Tools:** Git, Apache Kafka, Linux, Jenkins, JavaFX, Node.js, React, D3.js, Three.js

**Databases:** SQLite, MySQL, DynamoDB, FAISS, Pinecone

## EXPERIENCE

### Cox Automotive

Atlanta, GA

Software Engineering Intern

May 2025 - August 2025

- Designed and productionized SkeyLift, a Slack-based platform for automated API key provisioning, now actively used by 12+ internal engineering teams across Cox for secure access to platform services.
- Led end-to-end product and UI definition for SkeyLift, translating engineer pain points into interaction flows, Slack command UX, modal layouts, and permission states; iterated designs in Figma prior to implementation to clarify edge cases and reduce rework.
- Designed internal UX artifacts including flow diagrams, error-state behaviors, and system feedback patterns to align backend constraints with user-facing affordances.
- Built DynamoDB lifecycle storage with TTLs, IAM-scoped access, and CloudWatch logging to expose key expiry, requester identity, and provisioning metadata for auditing, debugging, and compliance reviews.
- Implemented request validation, permission checks, and failure handling to ensure reliable key issuance and prevent misuse across teams.
- Contributed to team deliverables beyond intern project, shipping features, bug fixes, and CI/CD improvements in a large enterprise monorepo.
- **Tech Stack:** *Python, AWS (Lambda, API Gateway, DynamoDB, S3), Terraform, Docker, Git, GitHub Actions, Slack Bolt API*

### Developmental Dynamics Lab UGA

Athens, GA

Data Visualization Researcher

August 2024 - Present

- Engineered a Java-based desktop application with optional web interface for real-time multimodal data visualization synchronized with video footage, improving processing efficiency by 80%.
- Designed UI layouts, interaction patterns, and visual hierarchy for multimodal datasets using Figma, then implemented them in React and D3.js balancing perceptual clarity, interaction latency, and scalability for dense time-series data.
- Built AI-powered data preprocessing pipelines using Python, MATLAB, and scikit-learn to smooth physiological signals and reduce sensor noise.
- Designed a scalable backend using Kafka for event streaming, Hibernate for persistence, and SQLite for offline access.
- Developed and automated testing frameworks using JUnit and Mockito for validating new features, integrating Jenkins for CI/CD.
- Integrated TLS 1.3 for data in transit and AES-256 for data at rest to secure data transmission and storage in all processing components.
- Engineered visual components in React/D3.js, optimizing SVG/Canvas rendering and interaction logic for large multimodal datasets.
- **Tech Stack:** *Java, Python, scikit-learn, MATLAB, Hibernate, Apache Kafka, SQLite, JUnit, Mockito, Jenkins, TLS 1.3, AES-256*

## PROJECTS

### KTPHacks 2025 (1st Place) - Eyewitness

November 2025

- Architected “Eyewitness,” an AI full-stack computer-vision application for real-time eye and head movement tracking and attention visualization.
- Designed UX and UI Figma prototypes for heatmaps, timelines, and attention summaries prior to implementation.
- Designed a Python backend using MediaPipe Face Mesh and EyeTrax to extract 3D eye-landmark data, compute gaze vectors at 30 FPS, and stream them through a processing layer built with OpenCV and NumPy for smoothing, blink detection, and noise filtering.
- Integrated Hugging Face API and Claude API to generate attention-region summaries and descriptive scene annotations from gaze trajectories.
- Implemented an interactive Next.js + Framer Motion frontend to visualize gaze analytics.
- **Tech Stack:** *Python, JavaScript, Node.js, Next.js, Claude API, Hugging Face API, MediaPipe, EyeTrax, OpenCV, NumPy, REST APIs, Framer Motion*

### Healthcare Accessibility Application

March 2025

- Designed and built a full-stack JavaFX application to automate healthcare cost analysis and provider discovery.
- Integrated an AI-assisted data preprocessing pipeline using Hugging Face transformer models and scikit-learn feature transformers to classify malformed API responses, infer missing provider attributes, and normalize structured fields before visualization.
- Added a high-throughput REST API layer with schema normalization and request batching to support 500+ unique user queries in real time.
- Reduced provider search time by ~40% with multithreaded request handling and adaptive throttling.
- **Tech Stack:** *Java, Hugging Face API, scikit-learn, JavaFX (FXML, CSS), REST APIs (CMS.gov, Ziptastic, NPPEs), HTTP/JSON, Maven, JUnit*